

Concept 5 | Advanced Theorems and Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Addition

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

Double angle

$$\sin 2A = 2 \sin A \cos A$$

Cosine

$$\cos 2A = \cos^2 A - \sin^2 A$$

Classified Practice

Advanced Theorems and Applications

Q001 Written

Find the maximum and minimum of $y = \sin^2\theta + \cos\theta + 1$, $0 \leq \theta < 2\pi$, and the corresponding θ .

Q002 Written**Find the maximum and minimum of $y = \cos^2\theta - \sin\theta$, $0 \leq \theta < 2\pi$, and the corresponding θ .**

Classified Practice

Advanced Theorems and Applications

Q003 Written

Solve the four source maximum/minimum problems on $0 \leq \theta < 2\pi$, including $y = \sin^2 \theta + \sin \theta - 1$, $y = \cos^2 \theta - \cos \theta + 1$, $y = 2 \sin \theta + \cos^2 \theta - 1$, and $y = -2 \sin^2 \theta - 2 \cos \theta + 1$. Include the source challenge $y = \cos^2 \theta - \sin \theta$.

Classified Practice

Advanced Theorems and Applications

Q004 **Written****Use addition theorems to find $\sin 75^\circ$ and $\cos(\pi/12)$.**

Q005 Written

Given $\sin \alpha = \frac{\sqrt{3}}{2}$ **in QII** and $\cos \beta = -\frac{3}{5}$ **in QIII**, **find** $\sin(\alpha + \beta)$ **and** $\cos(\alpha - \beta)$.

Classified Practice

Advanced Theorems and Applications

Q006 Written

Complete the source addition-theorem exercises *exact values, quadrant – controlled* α, β , and *the three – angle challenge* ($\sin \alpha = 0.6, \cos \beta = 0.8, \tan \gamma = 0.75$).

Q007 Written**Find $\tan 105^\circ$, then find the angle between $y = 2x + 1$ and $y = -3x + 2$.**

Classified Practice

Advanced Theorems and Applications

Q008 Written

Find the angle between each source pair of straight lines, including $y = \frac{1}{3}x$ with $y = 2x$, and $8x + 2y - 3 = 0$ with $-5x + 3y + 1 = 0$.

Classified Practice

Advanced Theorems and Applications

Q009 Written

Find the exact angles for all source line pairs and solve the challenge: a line through $P(2, 3)$ makes acute angle $\tan \theta = 3/4$ with $x + 2y - 4 = 0$; give both line equations and the areas of the triangles with the y -axis.

Classified Practice

Advanced Theorems and Applications

Q010 Written

Given $\sin \alpha = 1/3$ with α in QII, find $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$.

Q011 Written

Given $\sin \alpha = -2/3$ **with** $2\pi < \alpha < 3\pi$, **find** $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$.

Classified Practice

Advanced Theorems and Applications

Q012 **Written**

Solve the source double-angle exercises for given sine/cosine values and the challenge $\cos \theta = 0.6, \pi < \theta < 2\pi$, evaluating $\cos 3\theta \tan 2\theta$.

Q013 Written

Given $\cos \alpha = 3/5$, $0 < \alpha < \pi$, **find** $\sin(\alpha/2)$, $\cos(\alpha/2)$, $\tan(\alpha/2)$, **and evaluate** $\cos(5\pi/8)$.

Classified Practice

Advanced Theorems and Applications

Q014 **Written**

Given $\cos \alpha = -1/3$, $0 < \alpha < \pi$, **find** $\sin(\alpha/2)$, $\cos(\alpha/2)$, $\tan(\alpha/2)$, **and evaluate** $\sin(3\pi/8)$.

Q015 Written

Solve the source half-angle exercises for the three quadrant intervals and the challenge with $\tan A = \frac{\sqrt{5}}{2}, 4\pi < A < 5\pi,$
 $\cos B = -\frac{\sqrt{5}}{3}, 5\pi < B < 6\pi.$

Classified Practice

Advanced Theorems and Applications

Q016 **Written****Solve on** $0 \leq \theta < 2\pi$: $a \sin 2\theta + \sin \theta = 0$, $b \cos 2\theta - 7 \cos \theta + 4 \geq 0$.

Q017 Written**Solve the source equation** $\sin 2\theta + \cos \theta = 0$ **on** $0 \leq \theta < 2\pi$.

Classified Practice

Advanced Theorems and Applications

Q018 **Written**

Solve all source double-angle equations and inequalities on $0 \leq \theta < 2\pi$, including the challenge $\cos 2\theta - 7 \cos \theta + 4 \geq 0$.

Q019 Written

Rewrite $\sin \theta - \sqrt{3} \cos \theta$ as $r \sin(\theta + \alpha)$, $r > 0$, $-\pi < \alpha < \pi$.

Classified Practice

Advanced Theorems and Applications

Q020 Written

Rewrite the source expressions $\sin \theta - \cos \theta$, $\sin \theta + \cos \theta$, $3 \sin \theta - \sqrt{3} \cos \theta$, and $3 \sin \theta - \cos \theta$ in harmonic-addition form.

Q021 Written

Rewrite all source harmonic-addition exercises in the form $r \sin(\theta + \alpha)$, including $2 \sin \theta + 2 \cos \theta$ and the challenge $\sin \theta - \sqrt{3} \cos \theta$.

Classified Practice

Advanced Theorems and Applications

Q022 Written

Solve on $0 \leq \theta < 2\pi$: $a \sin \theta - \cos \theta = 1$, $b \sin \theta + \cos \theta < 3/2$.

Q023 Written**Solve** $\sin \theta \leq \cos \theta$ **on** $0 \leq \theta < 2\pi$.

Classified Practice

Advanced Theorems and Applications

Q024 Written

Solve all source harmonic-addition equations and inequalities on $0 \leq \theta < 2\pi$, including $\sin \theta + \cos \theta + 1 = 0$ and $\sin \theta - \cos \theta + 3/2 > 0$.

Q025 Written

Find the maximum and minimum of $y = 2 \sin(\theta - \pi/3)$, and of $y = 2 \cos 2\theta + 4 \cos \theta + 1$, $0 \leq \theta < 2\pi$.

Classified Practice

Advanced Theorems and Applications

Q026 Written

Find the maximum and minimum of the source functions $y = -\frac{4}{3} \sin^2 \theta + \cos \theta$ and $y = \cos 2\theta - 2 \cos \theta$.

Q027 Written

Solve all source maximum/minimum problems in the final lesson, including absolute-value and double-angle forms, and state the corresponding angles where requested.

Classified Practice

Advanced Theorems and Applications

Q028 MCQ

The substitution for $\sin^2 \theta + \cos \theta + 1$ should be:

Choose the correct option.

A $t = \sin \theta$

B $t = \cos \theta$

C $t = \tan \theta$

D $t = \theta$

Q029 MCQ**The maximum of $\sin^2 \theta + \cos \theta + 1$ is:****Choose the correct option.****A** 0**B** 1**C** 9/4**D** 4

Classified Practice

Advanced Theorems and Applications

Q030 MCQ

An extremal vertex is valid only if t lies in:**Choose the correct option.**A $[-2,2]$ B $[-1,1]$ C $[0,1]$

D all real numbers

Q031 MCQ

The identity for $\sin(a+b)$ is:**Choose the correct option.**

- A $\sin \alpha \cos \beta + \cos \alpha \sin \beta$
- B $\sin \alpha \sin \beta$
- C $\cos \alpha \cos \beta$
- D $\tan \alpha + \tan \beta$

Classified Practice

Advanced Theorems and Applications

Q032 MCQ

sin 75 degrees equals:**Choose the correct option.**A $(\sqrt{6}-\sqrt{2})/4$ B $(\sqrt{6}+\sqrt{2})/4$ C $1/2$ D $\sqrt{3}/2$

Q033 MCQ**A missing trig value is selected using:****Choose the correct option.**

- A the quadrant
- B the denominator
- C the period only
- D a graph scale

Classified Practice

Advanced Theorems and Applications

Q034 MCQ

The slope of $y=mx+k$ is:**Choose the correct option.**

A k

B m

C $1/m$ D $m+k$

Q035 MCQ**The angle formula uses denominator:****Choose the correct option.****A** $1+m_1m_2$ **B** m_1-m_2 **C** m_1+m_2 **D** m_1m_2

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Advanced Theorems and Applications

Q036 MCQ

For an acute angle, if $\tan \theta_1$ is negative we use:

Choose the correct option.

A θ_1

B $\pi - \theta_1$

C $2\theta_1$

D $-\theta_1$ only

Q037 MCQ **$\sin(2a)$ equals:****Choose the correct option.**

- A $\sin \alpha$
- B $2 \sin \alpha \cos \alpha$
- C $\cos^2 \alpha$
- D $\tan \alpha$

Classified Practice

Advanced Theorems and Applications

Q038 MCQ

 $\cos(2\alpha)$ can be written as:**Choose the correct option.**A $\cos^2 \alpha - \sin^2 \alpha$ B $2\sin \alpha$ C $\tan \alpha$ D $1 + \sin \alpha$

Q039 MCQ**For θ in QIV, $\sin \theta$ is:****Choose the correct option.**

- A positive
- B negative
- C zero always
- D undefined

Classified Practice

Advanced Theorems and Applications

Q040 MCQ

The half-angle formula for $\sin(a/2)$ uses:**Choose the correct option.**A $(1 - \cos \alpha)/2$ B $(1 + \cos \alpha)/2$ C $\tan \alpha$ D $\cos 2\alpha$

Q041 MCQ**The sign of a half-angle root is chosen from:****Choose the correct option.**

- A the original quadrant interval
- B the coefficient only
- C the denominator
- D the period of $\tan$

Classified Practice

Advanced Theorems and Applications

Q042 MCQ

If $\cos a = -1/3$ and $0 < a < \pi$, $\cos(a/2)$ is:

Choose the correct option.

A negative

B positive

C zero

D undefined

Q043 MCQ**For double-angle equations, first rewrite in:****Choose the correct option.**

- A one trig function
- B two unrelated angles
- C degrees only
- D a logarithm

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Advanced Theorems and Applications

Q044 MCQ

The factorisation of $\sin(2\theta) + \sin\theta$ is:**Choose the correct option.**A $\sin\theta(2\cos\theta + 1)$ B $\cos\theta(2\sin\theta - 1)$ C $\sin\theta + \cos\theta$ D $2\sin\theta$

Q045 MCQ**A tangent asymptote is always:****Choose the correct option.**

- A included
- B excluded
- C a maximum
- D an endpoint

Classified Practice

Advanced Theorems and Applications

Q046 MCQ

For $a \sin \theta + b \cos \theta$, r is:**Choose the correct option.**A $a+b$ B $\sqrt{a^2+b^2}$ C $a-b$ D ab

Q047 MCQ

The harmonic target form is:**Choose the correct option.**A $r \sin(\theta + \alpha)$ B $r \cos \theta$ C $a \tan \theta$ D $\sin \theta$ only

Classified Practice

Advanced Theorems and Applications

Q048 MCQ

The angle a is located from point:**Choose the correct option.**

- A $P(a,b)$
- B $P(r,\theta)$
- C the origin only
- D the x intercept

Q049 MCQ**After synthesis, t is usually:****Choose the correct option.****A** $\theta + \alpha$ **B** θ^2 **C** $\tan \theta$ **D** $\cos \theta$

Classified Practice

Advanced Theorems and Applications

Q050 MCQ

If $\max(\sin \theta + \cos \theta) = \sqrt{2}$, then $\sin \theta + \cos \theta < \frac{3}{2}$ is:

Choose the correct option.

- A never true
- B always true
- C true only at π
- D undefined

Q051 MCQ**In a strict inequality, equality endpoints are:****Choose the correct option.**

- A included
- B excluded
- C doubled
- D random

Classified Practice

Advanced Theorems and Applications

Q052 MCQ

The range of $2\sin(\theta - \pi/3)$ is:**Choose the correct option.**A $[-1,1]$ B $[-2,2]$ C $[0,2]$ D $[-3,3]$

Q053 MCQ**For a double-angle maximum problem, use:****Choose the correct option.**

- A the double-angle formula then a bounded variable
- B a logarithm
- C only a graph
- D no substitution

Classified Practice

Advanced Theorems and Applications

Q054 MCQ

A vertex outside $[-1,1]$ is:**Choose the correct option.**

- A an allowed extremum
- B rejected
- C always the maximum
- D always the minimum

Quiz MCQ 1 Concept Quiz · MCQ

The maximum of $2 \sin(\theta - \pi/3)$ is:

Choose the correct option.

A -2

B -1

C 1

D 2

Classified Practice

Advanced Theorems and Applications

Quiz MCQ 2 Concept Quiz · MCQ

The addition-theorem value $\sin 75^\circ$ is:

Choose the correct option.

A $(\sqrt{6} + \sqrt{2})/4$

B $(\sqrt{6} - \sqrt{2})/4$

C $1/2$

D 1

Quiz MCQ 3 Concept Quiz · MCQ

For $y = 2x^2 + 4x + 1$, the vertex x-value is:

Choose the correct option.

A -2

B -1

C 1

D 2

Classified Practice

Advanced Theorems and Applications

Quiz MCQ 4 **Concept Quiz · MCQ**

The angle between slopes 2 and -3 is:

Choose the correct option.

A $\pi/6$

B $\pi/4$

C $\pi/3$

D $\pi/2$

Quiz WRITTEN 5 Concept Quiz · Written

Find the maximum and minimum of $y = \sin^2 \theta + \cos \theta + 1$ on $0 \leq \theta < 2\pi$.

Classified Practice

Advanced Theorems and Applications

Quiz WRITTEN 6 Concept Quiz · Written

Find $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$ **when** $\sin \alpha = 1/3$ **and** α **is in** QII.

Quiz WRITTEN 7 Concept Quiz · Written

Rewrite $3 \sin \theta - \sqrt{3} \cos \theta$ **as** $r \sin(\theta + \alpha)$.

Classified Practice

Advanced Theorems and Applications

Quiz WRITTEN 8 Concept Quiz · Written**Solve** $\sin \theta \leq \cos \theta$ **on** $0 \leq \theta < 2\pi$.

Answer key

Use the question number shown on each page.

- Q001** $\max y = \frac{9}{4}$ at $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$; $\min y = 0$ at $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$
- Q002** $\max y = \frac{5}{4}$ at $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$; $\min y = -1$ at $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$
- Q003** See the numbered solution: substitute $t = \sin \theta$ or $t = \cos \theta$, complete the square, then test $t = \pm 1$.
- Q004** $\sin 75^\circ = \cos \pi/12 = \frac{\sqrt{6} + \sqrt{2}}{4}$
- Q005** $\sin \alpha + \beta = \frac{4 - 3\sqrt{3}}{10}$; $\cos \alpha - \beta = \frac{3 - 4\sqrt{3}}{10}$
- Q006** See the numbered solution; the three-angle challenge gives $\sin(\alpha + \beta + \gamma) = \frac{117}{125}$.
- Q007** $\tan 105^\circ = -2 + \sqrt{3}$; $\theta = \frac{\pi}{4}$
- Q008** The first pair gives $\theta = \frac{\pi}{4}$; the second pair also gives $\theta = \frac{\pi}{4}$.
- Q009** See the numbered solution; the two slopes are $m = 2/11, -2$, with triangle areas $49/165$ and $25/3$.
- Q010** $\sin 2\alpha = -\frac{4\sqrt{2}}{9}$, $\cos 2\alpha = \frac{7}{9}$, $\tan 2\alpha = -\frac{4\sqrt{2}}{7}$
- Q011** $\sin 2\alpha = \frac{4\sqrt{5}}{9}$, $\cos 2\alpha = \frac{1}{9}$, $\tan 2\alpha = 4\sqrt{5}$
- Q012** For the challenge, $\cos 3\theta = -\frac{117}{125}$, $\tan 2\theta = \frac{24}{7}$, so the product is $-\frac{2808}{875}$.
- Q013** $\sin \frac{\alpha}{2} = \frac{1}{\sqrt{5}}$, $\cos \frac{\alpha}{2} = \frac{2}{\sqrt{5}}$, $\tan \frac{\alpha}{2} = \frac{1}{2}$, $\cos \frac{5\pi}{8} = -\frac{\sqrt{2-\sqrt{2}}}{2}$
- Q014** $\sin \frac{\alpha}{2} = \sqrt{\frac{2}{3}}$, $\cos \frac{\alpha}{2} = \frac{1}{\sqrt{3}}$, $\tan \frac{\alpha}{2} = \sqrt{2}$, $\sin \frac{3\pi}{8} = \frac{\sqrt{2+\sqrt{2}}}{2}$
- Q015** Use the signed half-angle formulas; for the challenge, $\cos(A/2 - B/2) = \frac{\sqrt{3+\sqrt{5}} - \sqrt{15-5\sqrt{5}}}{6}$, $\sin(A/4) = \sqrt{\frac{1-\sqrt{5/6}}{2}}$.
- Q016** $a\theta = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}$; $b\frac{\pi}{3} \leq \theta \leq \frac{5\pi}{3}$
- Q017** $\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$
- Q018** See the numbered solution: replace double angles, solve in $s = \sin \theta$ or $c = \cos \theta$, and translate the accepted ranges.
- Q019** $r = 2$, $\alpha = -\frac{\pi}{3}$, $\sin \theta - \sqrt{3} \cos \theta = 2 \sin \theta - \frac{\pi}{3}$
- Q020** $\sqrt{2} \sin \theta - \frac{\pi}{4}$, $\sqrt{2} \sin \theta + \frac{\pi}{4}$, $2\sqrt{3} \sin \theta - \frac{\pi}{6}$, $\sqrt{10} \sin \theta - \arctan \frac{1}{3}$
- Q021** See the numbered solution; every result has $r = \sqrt{a^2 + b^2} > 0$ and α chosen in $(-\pi, \pi)$.
- Q022** $a\theta = \frac{\pi}{2}, \pi$; $b\ 0 \leq \theta < 2\pi$
- Q023** $0 \leq \theta \leq \frac{\pi}{4}$ or $\frac{5\pi}{4} \leq \theta < 2\pi$
- Q024** See the numbered solution; synthesize first, then solve in the shifted variable over its full range.
- Q040** A
- Q041** A
- Q042** B
- Q043** A
- Q044** A
- Q045** B
- Q046** B
- Q047** A
- Q048** A
- Q049** A
- Q050** B
- Q051** B
- Q052** B
- Q053** A
- Q054** B
- Quiz MCQ 1** D
- Quiz MCQ 2** A
- Quiz MCQ 3** A
- Quiz MCQ 4** B
- Quiz WRITTEN 5** $\max = \frac{9}{4}, \min = 0$
- Quiz WRITTEN 6** $-\frac{4\sqrt{2}}{9}, \frac{7}{9}, -\frac{4\sqrt{2}}{7}$
- Quiz WRITTEN 7** $2\sqrt{3} \sin \theta - \frac{\pi}{6}$
- Quiz WRITTEN 8** $[0, \frac{\pi}{4}] \cup [\frac{5\pi}{4}, 2\pi)$